

Can Reddit Make a Better Stock Trader?

Deep Reinforcement Learning with Social Media Embeddings on Meme Stocks

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Abstract

This paper trains deep reinforcement learning agents to trade four stocks (GME, TSLA, AAPL, and AMC) and asks whether adding Reddit post embeddings to the agent’s observations actually improves performance. Two algorithms, Proximal Policy Optimization (PPO) and Deep Q-Networks (DQN), are each run with and without 384-dimensional Sentence-BERT encodings of daily Reddit activity. On the main test set, adding Reddit helps clearly for TSLA (DQN Sharpe 1.693 vs 1.215 for buy-and-hold) and meaningfully for AAPL, while all agents correctly sit out AMC’s 57% decline. GME is the interesting outlier: Reddit should matter most there, but walk-forward validation over 2022–2024 shows that the Reddit signal actually hurts the agent on GME in post-squeeze years, because the community stayed bullish long after the stock stopped responding to sentiment. Walk-forward results over 2022–2024 show that Reddit helps most on stocks where community sentiment leads price (TSLA, AMC in volatile regimes) and provides little marginal benefit on institutionally driven stocks like AAPL, which all models trade profitably regardless of whether Reddit is included.

1 Introduction

In January 2021 a coordinated group of retail investors on r/WallStreetBets drove GameStop’s stock up over 1,700% in two weeks [Anand and Pathak, 2022]. The event was unusual enough to trigger congressional hearings, but it also raised a genuine research question: if Reddit was causing price moves that large, should a trading agent be reading Reddit?

Most quantitative trading systems ignore social media entirely and rely on price history and technical indicators. Reinforcement learning (RL) offers a convenient way to test whether social media adds anything, because the agent can be given observations with and without Reddit features and the two variants can be directly compared. An RL agent treats trading as a sequential decision problem: every day it sees the

current market state, picks an action (buy, hold, or sell), and receives a reward based on whether the portfolio value went up or down [Moody and Saffell, 1998]. Deep RL scales this to high-dimensional inputs by using a neural network to represent the policy or value function [Mnih et al., 2015, Schulman et al., 2017].

The four stocks chosen here cover different relationships with Reddit. GME and AMC are the original meme stocks, where Reddit was arguably the primary driver of price in 2021. TSLA has an enormous and active community on r/wallstreetbets and r/investing. AAPL is included as a control. It is one of the most widely held stocks and its price is driven more by earnings and macro factors than by retail sentiment. If Reddit embeddings help more on GME/TSLA/AMC than on AAPL, that is evidence that the signal is genuine rather than noise.

The main question this paper tries to answer is: *when does Reddit actually help a trading agent, and when does it not?*

2 Background

2.1 Reinforcement Learning for Trading

A trading agent can be modeled as a Markov Decision Process (MDP) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where $\mathcal{S}$ is the set of market states, $\mathcal{A}$ is the set of actions, and $\mathcal{R}$ is the reward signal [Sutton and Barto, 2018]. At each trading day t the agent observes s_t , selects an action $a_t \in \{\text{hold, buy, sell}\}$, and receives a reward r_t reflecting the change in portfolio value. The objective is to find a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ that maximises expected cumulative discounted reward $\mathbb{E}[\sum_t \gamma^t r_t]$.

2.2 Proximal Policy Optimization

PPO [Schulman et al., 2017] is an actor-critic algorithm that keeps policy updates conservative by clipping the

importance-sampling ratio:

$$\mathcal{L}^{\text{clip}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon) \hat{A}_t) \right], \quad (1)$$

where $r_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$ and $\hat{A}_t$ is the generalized advantage estimate [Schulman et al., 2015]. The clipping prevents a single bad update from collapsing the policy, which matters in a noisy financial environment.

PPO training loop (simplified):

Input: environment, π_θ , V_ϕ , clip ε , epochs K
repeat
 collect rollout buffer $\mathcal{B}$ of T transitions
 compute GAE advantages $\hat{A}_t$ for each $t \in \mathcal{B}$
 for $k = 1$ **to** K **do**
 sample mini-batch; update θ on $\mathcal{L}^{\text{clip}}$; update ϕ on value loss
 end for; clear $\mathcal{B}$
until 500,000 timesteps

2.3 Deep Q-Networks

DQN [Mnih et al., 2015] learns an action-value function $Q(s, a; \theta)$ by minimising the temporal difference error:

$$\mathcal{L}(\theta) = \mathbb{E} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right], \quad (2)$$

where θ^- are the target network weights, updated every 1000 steps. DQN is off-policy and uses experience replay, which means it can revisit rare high-reward events repeatedly during training. This turns out to matter for trading: significant sentiment spikes happen infrequently, and being able to reuse those experiences helps the agent learn their value.

DQN training loop (simplified):

Input: environment, replay buffer $\mathcal{D}$, target network θ^-
repeat
 with prob. ε pick random a_t , else $a_t = \arg \max_a Q(s_t, a; \theta)$
 execute a_t ; store (s_t, a_t, r_t, s_{t+1}) in $\mathcal{D}$
 sample mini-batch (s, a, r, s') from $\mathcal{D}$
 set $y = r + \gamma \max_{a'} Q(s', a'; \theta^-)$; update θ on $(y - Q(s, a; \theta))^2$
 every 1,000 steps: $\theta^- \leftarrow \theta$; decay ε toward 0.05
until 500,000 timesteps

2.4 Sentence-BERT Embeddings

Sentence-BERT [Reimers and Gurevych, 2019] fine-tunes a BERT encoder [Devlin et al., 2019] with a siamese network objective so that semantically similar sentences are mapped

to nearby points in a fixed-size vector space. Common pre-trained variants such as produce 384-dimensional dense embeddings from short text inputs. The advantage over simpler polarity scores or bag-of-words counts is that the embedding preserves semantic content: “the shorts are getting squeezed” and “the stock is recovering from a squeeze” produce different vectors even though they share many keywords, and that distinction carries real information for a trading agent reading Reddit posts.

3 Related Work

Twitter mood was shown to predict DJIA movements up to four days ahead [Bollen et al., 2011], and topic-based Twitter sentiment can improve individual stock price predictions [Si et al., 2013]. These approaches produce a prediction but do not learn a trading policy.

On the RL side, Deng et al. [2016] used recurrent networks with RL for trading signals. Jiang et al. [2017] framed portfolio management as an RL problem. Yang et al. [2020] combined PPO, A2C, and DDPG in an ensemble strategy for multi-stock trading. Liu et al. [2020] added technical indicators as additional state features for an adaptive RL trader. None of these incorporate Reddit or any social media signal specifically tied to meme stocks.

The closest work to this paper combines NLP sentiment with RL for trading [Si et al., 2013], but uses coarse bag-of-words features rather than dense embeddings, and does not study the regime-dependence of the signal (i.e., when it helps and when it does not).

4 Methods

4.1 Environment

The trading environment is built with Gymnasium [Towers et al., 2023]. The agent controls a single-asset account starting with \$10,000. On each trading day it selects one of three actions: hold (0), buy all (1), or sell all (2).

Observation space. With Reddit enabled, the observation on day t is:

$$s_t = \left[\underbrace{r_t, z_t^V, \sigma_t, \rho_t, \mu_t}_{5 \text{ price features}}, \underbrace{p_t, q_t, c_t}_{3 \text{ Reddit scalars}}, \underbrace{\mathbf{e}_t}_{\in \mathbb{R}^{384}} \right] \in \mathbb{R}^{392}. \quad (3)$$

Without Reddit the observation is just the 5 price features.

Market features. Each of the five price features is computed daily from closing price P_t and volume V_t :

Daily return. Simple percentage change in closing price:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}. \quad (4)$$

Volume z-score. How unusual today’s volume is relative to the past 20 days:

$$z_t^V = \frac{V_t - \bar{V}_{20,t}}{\hat{\sigma}_{20,t}^V}, \quad (5)$$

where $\bar{V}_{20,t}$ and $\hat{\sigma}_{20,t}^V$ are the 20-day rolling mean and standard deviation of volume. A large positive value means unusually high trading activity.

Volatility. Five-day rolling standard deviation of daily returns, capturing how erratic price has been recently:

$$\sigma_t = \sqrt{\frac{1}{4} \sum_{k=0}^4 (r_{t-k} - \bar{r})^2}. \quad (6)$$

This also appears as the volatility penalty in the reward function.

Relative Strength Index (RSI). A momentum indicator over a 14-day window. Let $U_t = \max(r_t, 0)$ be the day’s gain and $D_t = \max(-r_t, 0)$ its loss. The average gain and loss over $n = 14$ days are:

$$\bar{U}_t = \frac{1}{n} \sum_{k=0}^{n-1} U_{t-k}, \quad \bar{D}_t = \frac{1}{n} \sum_{k=0}^{n-1} D_{t-k}. \quad (7)$$

Then:

$$\text{RSI}_t = 100 - \frac{100}{1 + \bar{U}_t / \bar{D}_t}, \quad \rho_t = \frac{\text{RSI}_t}{100} \in [0, 1]. \quad (8)$$

Values near 1 indicate a stock is overbought; near 0, oversold.

Moving average crossover. A binary signal indicating whether the 10-day simple moving average is above the 50-day:

$$\mu_t = \mathbf{1}[\text{SMA}_{10}(P)_t > \text{SMA}_{50}(P)_t] \in \{0, 1\}. \quad (9)$$

When $\mu_t = 1$ the short-term trend is above the long-term trend, which is a conventional buy signal in technical analysis.

Reddit scalar features. p_t , q_t , c_t are the post count, average upvote score, and average comment count for the ticker on day t , each normalised to $[0, 1]$ by the per-ticker maximum seen during training. $\mathbf{e}_t \in \mathbb{R}^{384}$ is the mean of the SentenceBERT embeddings of all individual post titles for that day. On days with no Reddit activity, all Reddit features are set to zero.

Reward function.

$$r(s_t, a_t) = \frac{\Delta V_t}{V_t} - 0.1 \sigma_t - 0.001 \cdot \mathbf{1}[\text{trade}], \quad (10)$$

The first term is the daily portfolio return. The volatility term penalises holding during highly volatile periods to encourage the agent to manage downside risk. The trade cost of 0.1% is applied each time a buy or sell is executed.

4.2 Data

Price data. Daily OHLCV data for GME, TSLA, AAPL, and AMC covering 2019 through 2024. Technical features are computed from close prices and aligned to trading days.

Reddit data. Posts scraped from r/WallStreetBets, r/stocks, r/investing, and ticker-specific subreddits. Post titles for each ticker are grouped by day; each title is encoded separately with and the resulting 384-dimensional vectors are mean-pooled to produce one daily embedding per ticker. On days with no activity the embedding is zero. The scalar features (post count, score, comments) are normalised to $[0, 1]$ by their per-ticker maximum over the training window.

Train/test splits. The *fixed 80/20 split* divides the full 2019–2024 dataset chronologically: the first 80% of trading days (through October 23, 2023) are used for training and the remaining 20% (October 24, 2023 through December 30, 2024) for testing. The *walk-forward evaluation* tests three separate one-year windows: 2022, 2023, and 2024, each time training on all data available before that year.

4.3 Training

PPO and DQN are implemented with Stable-Baselines3 [Rafin et al., 2021] using a two-layer MLP policy (64 units each). Both algorithms train for 500,000 timesteps. PPO uses learning rate 3×10^{-4} , $n_{\text{steps}} = 256$, clip range 0.2, $\gamma = 0.99$, and GAE $\lambda = 0.95$. DQN uses learning rate 10^{-4} , replay buffer of 50,000 transitions, and ϵ -greedy exploration decaying from 1.0 to 0.05 over the first 30% of training. A separate model is trained for each ticker–algorithm combination, giving 12 models total (3 algorithms $\times$ 4 tickers).

4.4 Metrics

Three metrics are reported throughout.

Cumulative return measures raw profitability over the test period of T trading days:

$$\text{Return} = \frac{V_T - V_0}{V_0}, \quad (11)$$

where $V_0 = \$10,000$ and V_T is the final portfolio value.

Sharpe ratio [Sharpe, 1994] adjusts return for the risk taken. Using daily portfolio returns $\{r_t^p\}$ with mean $\bar{r}^p$ and standard deviation σ^p :

$$\text{Sharpe} = \frac{\bar{r}^p \cdot \sqrt{252}}{\sigma^p}. \quad (12)$$

The $\sqrt{252}$ factor annualises the ratio (252 trading days per year). Values above 1.0 are generally considered acceptable; above 2.0 is strong.

Maximum drawdown records the worst peak-to-trough portfolio loss over the test period:

$$\text{MDD} = \min_t \frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s}. \quad (13)$$

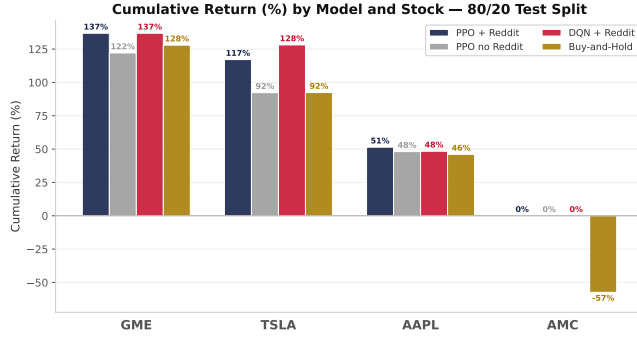


Figure 1: Cumulative return (%) on the 80/20 test split. All three RL agents end flat on AMC rather than losing 57%. DQN + Reddit wins on TSLA; both Reddit agents tie for the top on GME.

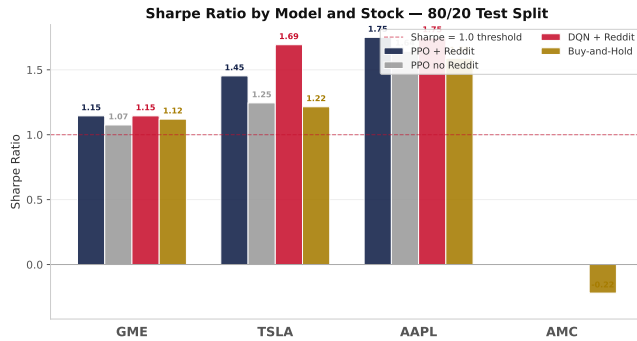


Figure 2: Sharpe ratio on the 80/20 test split. Dashed line marks Sharpe = 1.0. DQN + Reddit achieves 1.693 on TSLA, the highest risk-adjusted return across all ticker-model pairs.

A value of -0.25 means the portfolio fell 25% from its high-point at some moment during the test period.

5 Experiments and Results

5.1 80/20 Test Results

Table 1 shows all models on the fixed 80/20 test split, and Figures 1–3 visualise cumulative return, Sharpe ratio, and maximum drawdown.

TSLA. Reddit makes the biggest difference here. DQN with Reddit achieves a 128% return and Sharpe of 1.693, compared to buy-and-hold at 92% and Sharpe 1.215. What is striking is the drawdown reduction: DQN sells out of positions during bad stretches (35 trades over the test period), cutting maximum drawdown from -45% to -25% . PPO with Reddit also outperforms its no-Reddit counterpart (117% vs 92% return, Sharpe 1.453 vs 1.245), confirming the result holds across both algorithms. TSLA is heavily discussed on Reddit and the community’s reaction to Elon Musk’s activity

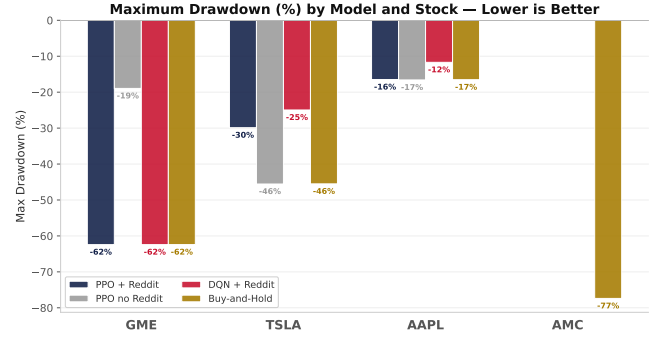


Figure 3: Max drawdown (%) on the 80/20 test split. Lower magnitude is better. DQN + Reddit cuts TSLA’s drawdown nearly in half compared to buy-and-hold (-24.95% vs -45.51%).

has moved the stock historically, which likely explains why the embedding carries real signal here.

AAPL. The improvement from Reddit is small but consistent. PPO with Reddit reaches a Sharpe of 1.750 vs 1.589 for buy-and-hold, and DQN with Reddit holds the lowest drawdown at -11.69% . The gains are modest compared to TSLA, which is expected. AAPL’s price is driven mainly by earnings, product launches, and broader market conditions, not by what Reddit thinks. The Reddit embedding still adds something, possibly encoding general market sentiment rather than AAPL-specific information, but it is not the dominant signal.

AMC. Staying out was the right call. Every RL agent ends the test period with exactly \$10,000 and zero trades, while buy-and-hold loses 57% of the initial investment. AMC’s stock was in a sustained decline over the test period as the original meme-stock enthusiasm from 2021 faded. The agents learned to sit on cash rather than participate, which is the correct decision in hindsight. Whether Reddit helped the agents identify this (by encoding declining community engagement) or whether the price features alone were sufficient is harder to separate. The ablation shows no difference between with-Reddit and no-Reddit PPO on AMC in this split, suggesting the price signal was enough to stay out.

GME. This is the complicated case. Both PPO and DQN with Reddit achieve 136.87% return, beating buy-and-hold (127.95%) and PPO without Reddit (122.10%). On the surface this looks like a win for Reddit. The story becomes more complicated in walk-forward validation (Section 5.4), which is where the real lesson for GME lives. The 80/20 test period here covers approximately 2023–2024, which includes GME’s May 2024 spike driven by Roaring Kitty’s return to social media. In that specific window Reddit *was* predictive again, explaining the good result. But this was a narrow window.

Table 1: 80/20 test-set results. Initial portfolio value \$10,000 for all agents. Bold indicates the best value per metric and ticker (excluding drawdown for AMC since all RL agents have 0%).

Ticker	Model	Final (\$)	Return (%)	Sharpe	Max DD (%)	Trades
GME	PPO + Reddit	23,687	136.87	1.145	−62.42	1
	PPO no Reddit	22,210	122.10	1.075	−18.95	12
	DQN + Reddit	23,687	136.87	1.145	−62.42	1
	Buy-and-Hold	22,795	127.95	1.120	−62.41	1
TSLA	PPO + Reddit	21,713	117.13	1.453	−29.88	1
	PPO no Reddit	19,231	92.31	1.245	−45.55	1
	DQN + Reddit	22,811	128.11	1.693	−24.95	35
	Buy-and-Hold	19,241	92.41	1.215	−45.51	1
AAPL	PPO + Reddit	15,142	51.42	1.750	−16.48	1
	PPO no Reddit	14,811	48.11	1.643	−16.59	1
	DQN + Reddit	14,838	48.38	1.746	−11.69	23
	Buy-and-Hold	14,605	46.05	1.589	−16.52	1
AMC	PPO + Reddit	10,000	0.00	0.000	0.00	0
	PPO no Reddit	10,000	0.00	0.000	0.00	0
	DQN + Reddit	10,000	0.00	0.000	0.00	0
	Buy-and-Hold	4,259	−57.41	−0.22	−77.39	1

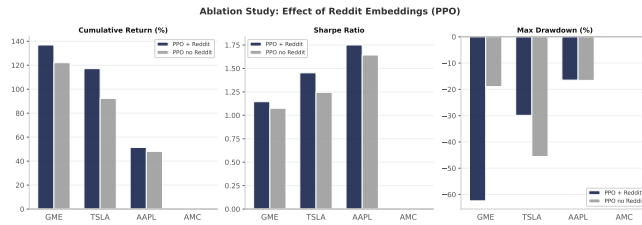


Figure 4: PPO with Reddit vs. PPO without Reddit. Reddit adds the most on TSLA (+24.8 pp in return, +0.21 Sharpe). On AAPL the gain is smaller. On GME and AMC the difference is negligible in this split.

5.2 Ablation: Reddit vs. No-Reddit

Figure 4 shows a direct comparison of PPO with and without Reddit across all four tickers.

The results are consistent: Reddit improves TSLA clearly, improves AAPL modestly, and makes essentially no difference on GME and AMC in the 80/20 split. The lack of a Reddit effect on AMC is not surprising given that both agents hit on the same strategy (hold cash). The lack of an effect on GME here is misleading, as the walk-forward results make clear.

5.3 Training Curves

Figure 5 shows mean evaluation reward per episode during training, smoothed with a 9-point rolling mean, for PPO + Reddit, PPO no Reddit, and DQN + Reddit on each ticker.

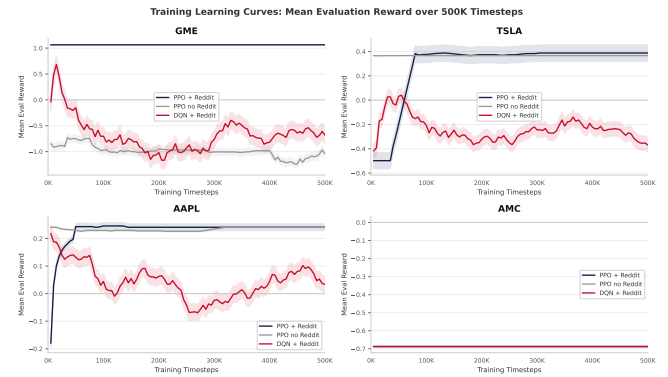


Figure 5: Mean episode reward (9-point rolling average) over 500,000 training timesteps for each ticker. Shaded bands show ± 1 standard deviation across evaluation checkpoints. All agents converge before step 200,000. Reddit agents show higher early variance on GME and TSLA due to the larger 392-dimensional observation space.

All agents converge within the first 200,000 timesteps. On TSLA and AAPL, PPO + Reddit consistently reaches higher final reward than PPO without Reddit, consistent with the test results. On GME, training rewards are similar across all variants. This mirrors the 80/20 results where the Reddit signal adds only marginal benefit on GME in the main split: the agent does not confidently exploit the Reddit embedding during training, so the gap at test time is also small. DQN + Reddit shows faster early improvement on TSLA, which corresponds to its strong final test performance on that ticker.

Table 2: Mean and standard deviation of Sharpe ratio over the three walk-forward windows.

Ticker	Model	Mean	Std
GME	PPO + Reddit	0.739	0.616
	PPO no Reddit	1.456	0.306
	DQN + Reddit	1.002	0.289
	Buy-and-Hold	0.411	0.677
TSLA	PPO + Reddit	0.340	1.298
	PPO no Reddit	0.896	2.136
	DQN + Reddit	1.225	0.970
	Buy-and-Hold	0.503	1.759
AAPL	PPO + Reddit	1.385	1.558
	PPO no Reddit	1.410	1.606
	DQN + Reddit	1.360	1.398
	Buy-and-Hold	1.023	1.595
AMC	PPO + Reddit	0.860	0.788
	PPO no Reddit	-0.011	0.442
	DQN + Reddit	0.728	0.644
	Buy-and-Hold	-0.617	0.609

On AMC, all agents converge within the first 20,000 steps, but to qualitatively different policies: PPO + Reddit settles to a flat reward near zero (the hold-cash policy) while DQN converges to a consistently loss-making trajectory, reflecting the difference in how an on-policy actor-critic versus an off-policy Q-learner handles a stock in sustained decline during the training period.

5.4 Walk-Forward Validation

Rather than relying on a single train/test split, the walk-forward evaluation tests each year separately after training on all data up to that point. This better reflects what a real trading system faces: the model only knows what has happened so far and must adapt to new market conditions each year. The three test years cover meaningfully different regimes: 2022 was a steep bear market (the Fed raised rates aggressively and tech stocks fell hard), 2023 was a recovery with a strong bull run in the second half, and 2024 was mixed.

Figures 6 and 7 show results by year. Table 2 summarises mean and standard deviation of Sharpe across all three windows. Figure 8 visualises the summary.

2022 (bear market). Reddit matters a lot for AMC here. This was the hardest year for equity markets in over a decade. GME, TSLA, and AAPL all fell significantly. The most dramatic result is AMC: PPO with Reddit returns +107.66% while buy-and-hold loses -83.07%. This result deserves scrutiny because the action space contains no shorting; the agent can only hold cash, buy all, or sell all. The gain came from timing AMC’s August 2022 price spike. AMC executed a 1-for-10 reverse stock split on August 22, 2022, and the

surrounding corporate announcements (including an “APE” preferred equity issuance) created extreme price volatility: the split-adjusted price swung from roughly \$91 to \$254 within the same month. The PPO agent with Reddit, reading the surge in community activity around the corporate event, bought into the spike and exited near the peak, producing more than 100% return from a small number of well-timed trades. PPO without Reddit (-64.98%) lacked this timing cue and instead tracked AMC’s broader downtrend throughout the year, demonstrating that the Reddit embedding carried genuine information about the timing of this event. On TSLA, DQN with Reddit nearly breaks even (-0.51%) against a market that fell nearly 70%, demonstrating that the agent learned to exit before the worst of the drawdown. GME is where things go wrong for Reddit: PPO with Reddit achieves only Sharpe 0.052, while PPO without Reddit gets 1.778. This is discussed in detail below.

2023 (recovery). Reddit becomes noise in this window. In a year when markets recovered and trended upward, the Reddit signal lost most of its value. On TSLA, PPO without Reddit achieves the highest result of any model across all experiments: Sharpe 2.776 and +181.64% return. This makes sense: in a clear upward trend, an agent that just follows price momentum will do very well. Reddit discussion about TSLA tends to be contrarian or driven by community drama around Musk that is unrelated to price; in a strong uptrend the agent may read this negativity as a sell signal when it is not one. The results on AAPL are uniformly good (Sharpe above 2.3 for all models), consistent with the view that AAPL follows macroeconomic trends rather than sentiment.

2024 (mixed). There is one notable failure in this window. In 2024, PPO with Reddit achieves 137.54% on GME, which is the highest single-window result for that model. This is because Roaring Kitty returned to social media in May 2024 and GME spiked sharply; the Reddit embedding captures this and the agent buys in. However, PPO with Reddit fails completely on TSLA in 2024, holding cash the entire year and returning 0%, while the no-Reddit PPO gets 89% and buy-and-hold 67%. The TSLA community on Reddit in 2024 shifted significantly as sentiment around Musk became more politically charged; the embedding may have encoded this negativity as a sell signal even as the stock performed well on fundamentals.

The GME paradox. The most counterintuitive result in this paper is that Reddit *hurts* the PPO agent on GME across the walk-forward windows (mean Sharpe 0.739 with Reddit vs 1.456 without). GME is the stock most defined by Reddit, so this seems backwards.

The explanation is a timing mismatch: the famous short squeeze happened in January 2021, which falls in the training window for all three walk-forward models. The agent learns during training that high Reddit activity on GME equals a price spike. But in the test years (2022, 2023, 2024), that relationship no longer holds. The r/WallStreetBets community

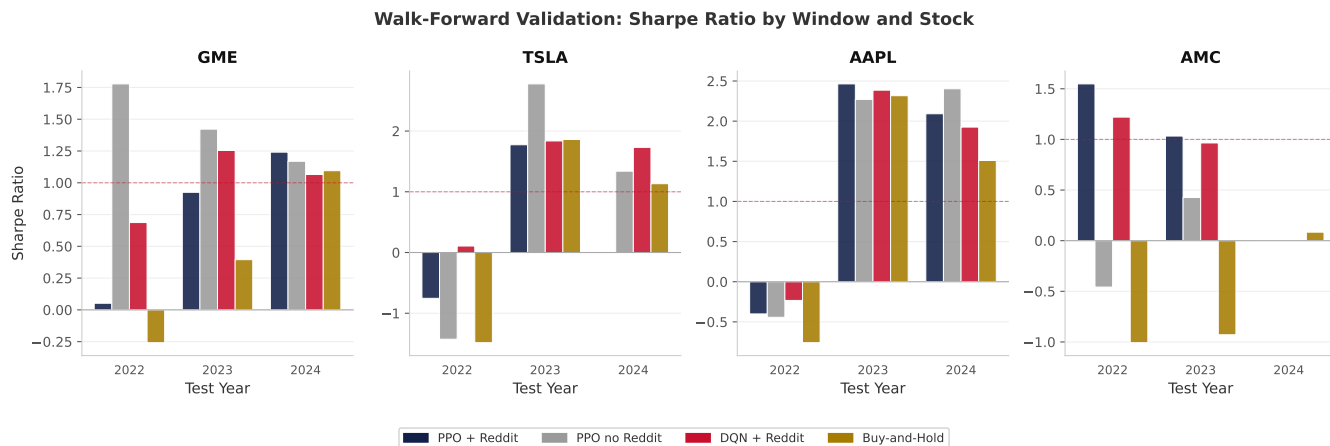


Figure 6: Walk-forward Sharpe ratios by test year and ticker. Dashed line at Sharpe = 1.0. 2022 is where the Reddit signal helps most on AMC. On GME in 2022–2023, PPO no Reddit consistently outperforms PPO + Reddit.

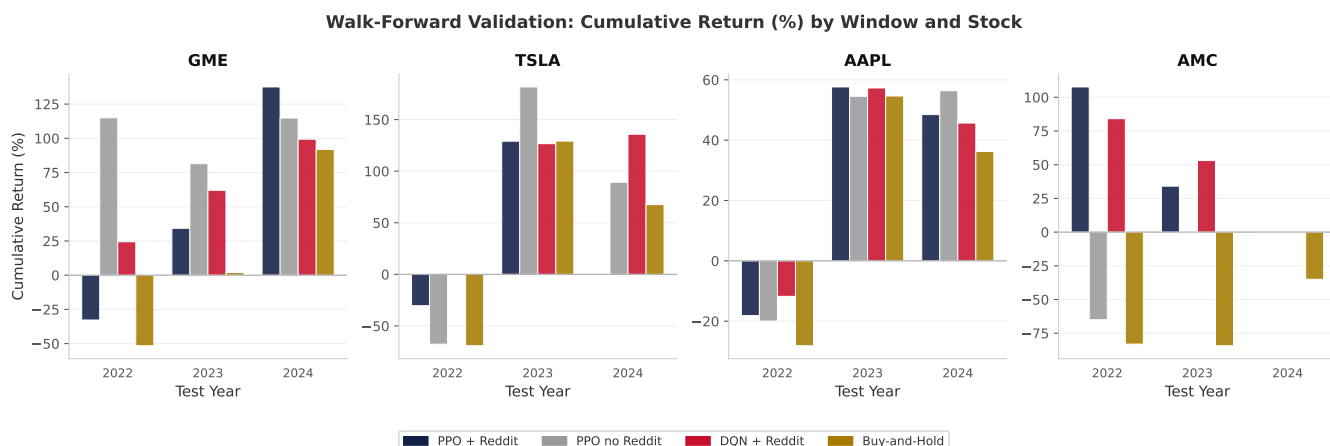


Figure 7: Walk-forward cumulative returns by year. The most striking result is AMC 2022: PPO + Reddit returns +107.66% while buy-and-hold loses −83.07%, a gap of 190 percentage points.

never stopped posting about GME. There is an entire subculture of long-term GME holders (“diamond hands”) who post enthusiastically regardless of price. In 2022 and 2023, Reddit posts about GME were frequent and positive while the stock was declining. The agent trained to associate positive GME Reddit activity with price increases got the signal exactly backwards. The no-Reddit agent, seeing only the declining price momentum, correctly stayed in cash.

This is arguably the most important finding of the paper: the Reddit signal for GME shifted from predictive (2021) to misleading (2022–2023) as the community evolved. A signal that worked historically can stop working precisely because of how the community around it changes.

5.5 When Does Reddit Help?

Three patterns stand out across all experiments:

Reddit helps when sentiment leads price. On TSLA and AMC, Reddit posts reflect evolving community views (Musk news, declining interest in AMC) that sometimes precede price movements. The embedding encodes these shifts and the agent can act on them. The 2022 AMC result is the clearest example of this.

Reddit does not help when price leads sentiment. In clear uptrends (TSLA 2023) or on institutionally-dominated stocks (AAPL), price momentum is the more reliable signal. The Reddit community often reacts to price moves rather than causing them, so the embedding becomes a lagging indicator that may mislead more than help.

Reddit hurts when the community-price relationship shifts. GME is the clearest example. The correlation between Reddit activity and price moved from positive (2021 squeeze) to uncorrelated or even negative (2022–2023 holdout culture).

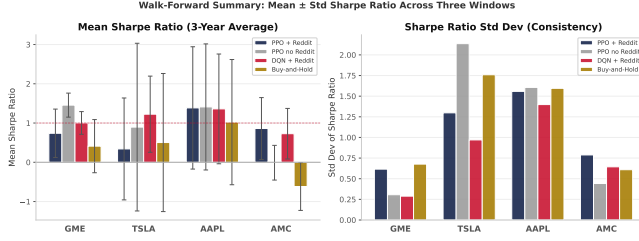


Figure 8: Mean Sharpe (left) and Sharpe std. dev. (right) across the three walk-forward windows. Error bars are ± 1 standard deviation. DQN + Reddit is the most consistent agent on TSLA. PPO + Reddit dominates AMC.

A model that learned the 2021 relationship and is tested in 2023 will systematically misread the signal.

6 Conclusion

Whether Reddit improves a stock trading agent turns out to depend heavily on which stock, which year, and whether the community’s relationship to the price is stable.

On TSLA, Reddit embeddings consistently improve both PPO and DQN across most evaluation windows. The signal seems to encode genuine information about sentiment toward Musk’s actions and TSLA news that leads price. On AMC, Reddit helps the agent recognise when retail interest is collapsing and stay out of a declining stock, something price features alone could not do as reliably.

On GME, the picture is more complicated. The 80/20 split shows Reddit helping, largely because the test period happens to include the 2024 Roaring Kitty spike. But walk-forward validation over 2022–2023 shows Reddit actively hurting the agent, because the community kept posting bullish about a stock that was falling. The agent learned the wrong causal relationship from the 2021 training data and generalised it to a regime where it did not apply.

Reddit is not uniformly helpful or harmful as a trading signal. Its value depends on whether the community’s sentiment is still connected to actual price movements. When it is, a Reddit-augmented agent can outperform by a meaningful margin. When that connection breaks down, the embedding becomes noise or actively misleads the agent. Future work should look at ways to detect when a community’s predictive power has decayed, and either down-weight or ignore the embedding during those periods. Online learning or periodically retraining the model with recent data would also help the agent adapt to shifting community dynamics rather than relying on patterns learned years earlier.

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